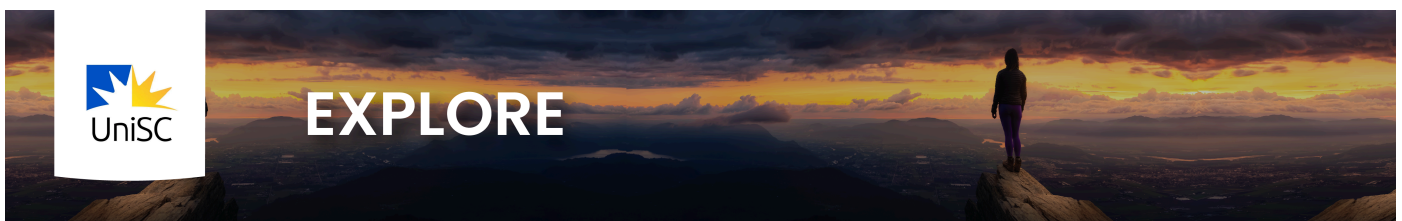


4.1 What is CAPM? How is beta estimated in practice? ↕



□ Guiding Questions

1. When you make an investment, what are the key reasons for expecting a return from it?
2. How does the CAPM combine these reasons into a single, coherent model?
3. What is the compensation due to time value of money in the CAPM?
4. What is the compensation due to risk called in the CAPM?
5. How is the CAPM beta estimated in practice?
6. How is the CAPM beta coefficient interpreted?
7. How can you estimate the CAPM beta in R?



What will you learn?

You will be introduced to the Capital Asset Pricing Model (CAPM) . You'll learn:

- How to write down the equation for the CAPM.
- The meaning of each term in the CAPM

- How to estimate the beta coefficient.
- How to interpret the results of estimation.

Motivation for the CAPM

When you make an investment, you expect a return from it. Why? There are two reasons:

- 1) Time value of money.
- 2) Risk

CAPM combines the above two reasons into a single, coherent, easy to understand form:

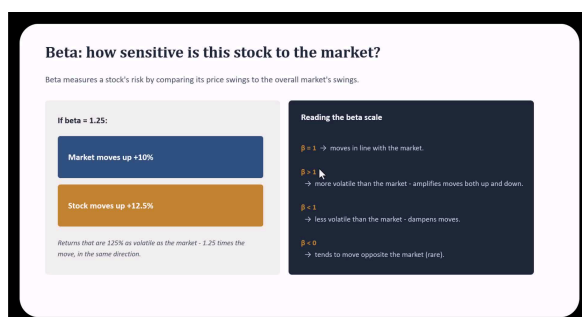
Expected return from an asset = compensation due to time value of money + compensation due to risk.

The compensation due to time value of money is the risk-free rate. The risk-free rate is typically taken to be the yield on a long-term (10 years) government bond.

The compensation due to risk is called the risk premium. Unsurprisingly, investors desire a greater rate of return when they take on more risk. In the CAPM, the risk-premium on a security is based on the beta of that security (this is the same beta that you encountered in module 3).

Watch

The Capital Asset Pricing Model



Details

Transcript

Search in transcript

1:50 is the market risk premium. The excess return

1:54 investors require, on average, to hold the market

1:58 portfolio rather than the risk free asset beta.

2:02 This term beta then scales this market wide risk

The CAPM Formula and Example Calculation

The CAPM formula is as follows:

$$E[R_s] = R_F + \beta_s \{E[R_M] - R_F\}$$

where:

$E[R_s]$ is the expected return on the security

$E[R_M]$ is the expected return on the market (typically proxied by a large broad-based index of the relevant market such as the S&P 500 index for the US stocks).

R_F is the risk-free rate.

β_s is the beta of the stock. It is a measure of a stock's risk calculated by measuring the fluctuation of its price changes relative to the overall market. In other words, it measures how sensitive the stock is to market risk. For instance, if a company's beta is equal to 1.25, it has returns that are 125% as volatile as the market it's being compared to. That is, if the market goes up by 10%, the company's stock is expected to go up by 12.5%. On the contrary, if the market goes down by 10%, the company's stock is expected to go down by 12.5%.

The CAPM formula is typically written as:

$$E[R_s] - R_F = \beta_s \{E[R_M - R_F]\}$$

where:

$E[R_s] - R_F$ is the excess return on the security

$E[R_M] - R_F$ is called the market risk-premium.

Let's put the above formula to use with the help of an example calculation. Suppose you are given the following information about a stock:

- It trades on the Australian Securities Exchange and its operations are based in Australia.
- Current yield on a Australian govt. 10-year treasury bond is 5%.
- The average excess historical annual return or market risk premium for Australian stocks is 8%
- The beta of the stock is 1.25 (meaning its average return is 1.25x as volatile as the ASX-200 over the last 2 years)

What is the expected return of the security using the CAPM formula?

Let's break down the answer using the above formula::

- Expected return = Risk Free Rate + [Beta x Market Return Premium]
- Expected return = 5% + [1.25 x 8%]
- Expected return = 15%

How is beta estimated in practice?

In practice, beta of a security is estimated by measuring a security's sensitivity to market movements via ordinary least square (OLS) regression. It is defined as the slope of the linear regression line where the dependent variable is the stock's excess return and the independent variable is the market's excess return (e.g., S&P 500 for the US stocks or ASX 200 for the Australian stocks).

The regression equation is:

$$R_s - R_F = \alpha + \beta(R_M - R_F) + \epsilon$$

where:

$R_s - R_F$ is the stock's excess return (Y variable)

$R_M - R_F$ is the market's excess return (X variable)

α is the intercept (abnormal return)

ϵ is the error term (residuals)

β is slope (systematic risk)

Step by Step Hypothetical Example:

Note that the following is a hypothetical example just for illustration purposes.

Suppose you are interested in estimating the beta of APPLE (AAPL) vs the S&P 500 by using 5 years monthly historical data (60 months). You look at the 10-year US treasury yield and find that it is approximately 4.3% per annum, which corresponds to 0.36% monthly. (It is common professional practice to just use the 10-year bond yield as the risk-free rate). The following steps are needed to estimate beta:

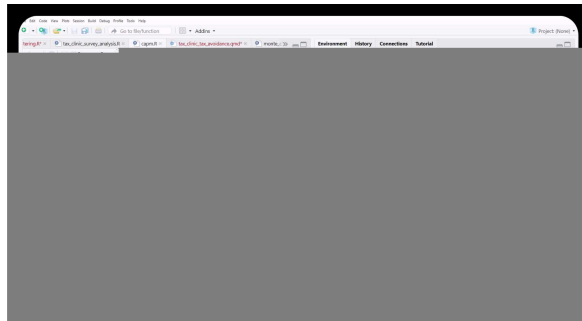
Step 1: Calculate monthly excess stock returns and monthly excess S&P 500 index returns.

Step 2: Run the OLS regression.

Step 3: Interpret the results. If beta is 1.2, then this tells you that for every 1% move in the market, the Apple stock is expected to move by 1.2%. If 'alpha' is positive, the stock outperformed the CAPM prediction. R-Squared tells you how much of the market's movement is explained by the market. Let's say R-Squared is 0.65. This means that 65% of the stock's movement is explained by the market.



Estimating the Capital Asset Pricing Model in R



Details

Transcript

 Search in transcript

- 0:00 Now let us walk through how to estimate the
- 0:03 capital asset pricing model and are using real
- 0:07 data. Please load the read Excel and the tidy
- 0:11 verse packages I'm using. Read Excel because I

<https://learn.usc.edu.au/courses/35359/files/3226152?wrap=1>

R Script <https://learn.usc.edu.au/courses/35359/files/3226152?wrap=1> [↓](#)
https://learn.usc.edu.au/courses/35359/files/3226152/download?download_frd=1

Now Let's Use Real Data: R Code for Beta Estimation

Download this [dataset](https://learn.usc.edu.au/courses/35359/files/3121378?wrap=1). https://learn.usc.edu.au/courses/35359/files/3121378/download?download_frd=1 [↓](#)
 You can download the R Script provided above.

Use the `read_excel` function in the `readxl` package. Install the packages as follows.

```
install.packages("readxl")
install.packages("dplyr")
```

Then, load the packages

```
library(readxl)
library(dplyr)
```

```
data <- read_excel("provide your file path here")
```

Build the dataset for estimating the CAPM model.

```
capm_data <- data %>%
  arrange(date) %>%
  mutate(
    # Convert annual % yield → monthly decimal risk-free rate
    rf = (tbill_10 / 100) / 12,

    # Monthly returns
    aapl_ret = aapl / lag(aapl) - 1,
    sp500_ret = sp500 / lag(sp500) - 1,

    # Excess returns
    aapl_excess = aapl_ret - rf,
    sp500_excess = sp500_ret - rf
  )
```

Estimate the CAPM Model

```
model <- lm(aapl_excess ~ sp500_excess, data = capm_data)
summary(model)
```

Interpretation of your CAPM results

Beta = 1.094

Apple is:

- **9.4% more volatile** than the market
- Moves slightly more than 1-for-1 with the S&P500
- A typical high-beta tech stock

Alpha = 0.00308 (not significant)

- Monthly alpha $\approx$ **0.31%**
- p-value = **0.665** → statistically indistinguishable from zero
- Apple did **not** outperform the market after adjusting for risk

R² = 0.452

- About **45%** of Apple's excess returns are explained by the market
 - The rest is firm-specific (earnings, product cycles, news, etc.)
-

Critically evaluate

Two assets both report a 15% return over the past year. Does it automatically follow that they are equivalent in their performance?